

Architecture Overview

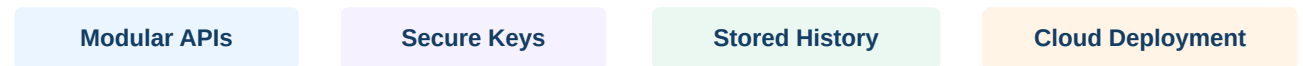
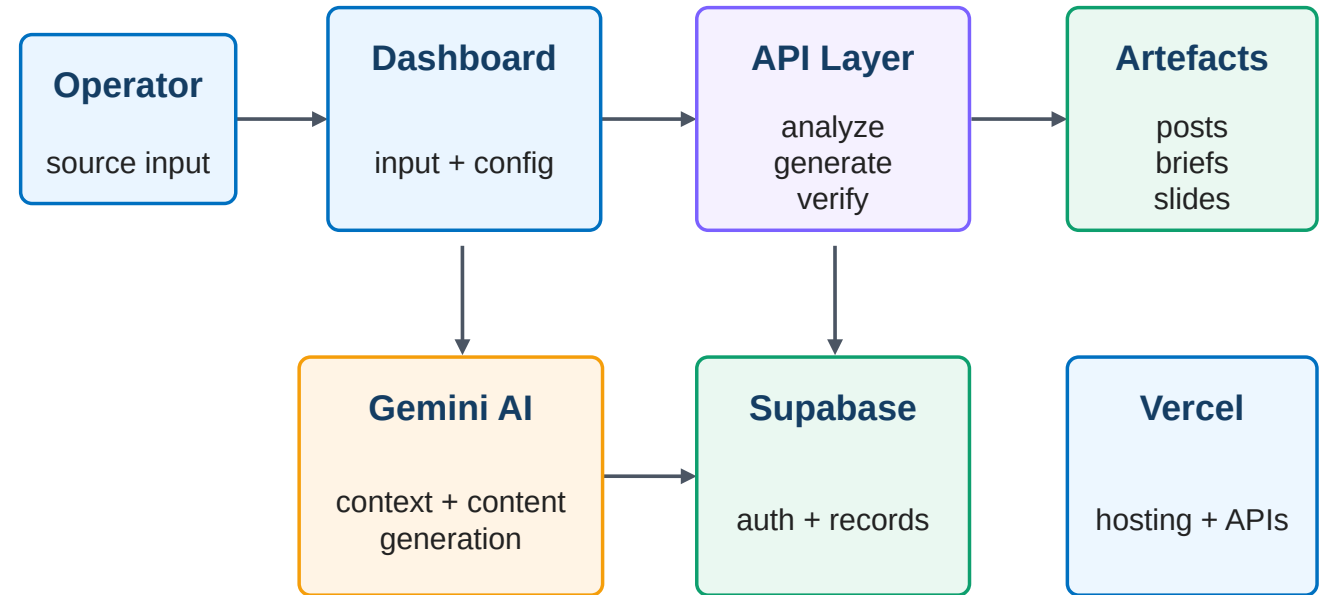
Transform AI is a web-based content transformation platform. The operator submits source content, selects required output formats and receives AI-generated communication artefacts through a simple dashboard.

Core Stack

- ◆ **Frontend** index.html dashboard and controls
- ◆ **APIs** Vercel serverless functions
- ◆ **AI Layer** Gemini analysis and generation
- ◆ **Data Layer** Supabase Auth + Postgres records

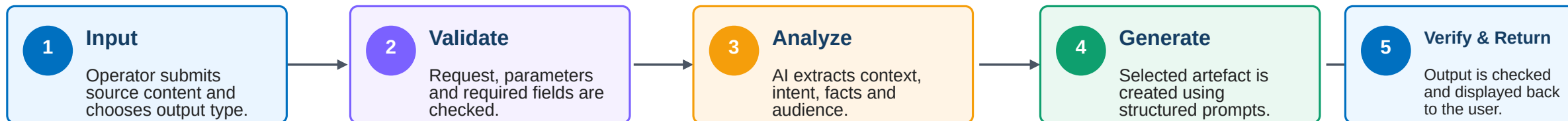
Design Principle

Keep the UI simple, API logic modular and secret keys protected on the backend.



Browser UI → Serverless APIs → AI Engine → Verified Output

Request Flow



Operational Modules

API LAYER

- analyze.js: extracts context and intent
- generate.js: creates selected artefacts
- verify.js: checks relevance and format

DATA & SECURITY

- Supabase Auth manages access
- Postgres stores user records and outputs
- API keys remain in environment variables

SCALABILITY

- Vercel serverless APIs scale per request
- New output types can be added as modules
- Database supports future audit/history

Supported Artefacts

Video Package • LinkedIn Post • Twitter/X Thread • Advisory • Infographic • Executive Summary • Presentation

Outcome: one common input can produce multiple structured communication outputs with reduced manual effort and improved consistency.